Elastic Bounds of Pearson's r: Theoretical and Empirical Insights for Ordinal Variables

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Abstract

The use of product-moment correlations to assess relationships between ordered-categorical variables is widespread and generally tolerated in spite of the non-metric character of such variables. Potential problems of this practice have been identified in the literature. This article focusses on one of these, namely that correlation coefficients are poorly comparable across pairs of ordered-categorical variables because their ranges are constrained in non-uniform ways. This not only affects the validity of correlation coefficients for descriptive purposes, but also as bases for subsequent analyses, as in, e.g., principal component analysis, factor analysis, and structural equation modelling. This article focusses on this problem of 'constrained' correlations. It explains the problem and explores conditions under which it is most severe. It provides illustrations based on both fabricated and actual empirical data, the latter from a mass-survey that ubiquitously contains many ordered-categorical variables that are commonly analysed with product-moment correlations. The article is accompanied by a software tool in R for analysing the attainable upper- and lower bounds of correlations in actual data.

Pearson's correlation coefficient r, when applied to ordinal data, is subject to elasticity due to constraints from marginal distributions. This paper develops theoretical bounds for r, based on Boole–Fréchet–Hoeffding inequalities and rearrangement inequalities, and derives analytic expressions for their limits. We investigate when symmetry in the bounds holds or breaks, and explore practical implications for correlation-based methods like PCA and SEM. An accompanying R package allows users to compute bounds and perform hypothesis tests given fixed marginals.

1 Introduction and Conceptual Setting

In this article we elaborate why correlations involving ordered-categorical variables are generally 'constrained', which means that their attainable maximum and minimum values are rarely +1 and -1, as is often thought, but values that have to be calculated for each pair of variables. We assess the magnitude of these constraints in fictitious as well as empirical data and analyse under which conditions they are most severe. The article is accompanied by a software tool in R: that allows applied researchers to easily assess the extent of the problem in their own data.

The use of product-moment correlations (also known as Pearson correlations) to assess linear relationships between ordered-categorical variables –henceforth referred to as orcats– is widespread and widely condoned in empirical studies, in spite of such variables not being metric while the product-moment correlation assumes that they are. We do not focus on this problem from an obsessive desire for methodological purity, were it only because many statistical procedures are quite robust for violations of their assumptions [fn: examples?]. Instead, we draw attention to problematic consequences of this violation of assumptions that, as far as we know, have been rarely recognised in the applied literature although they are well-known in the theoretical statistical literature. This is the problem that correlation coefficients involving orcats are constrained in their range which makes them mutually incomparable, leading to problems in hypothesis testing and in analytical procedures that use correlations as their input.

Despite long-standing cautions about applying Pearson's correlation coefficient r to ordinal variables ([**Labovitz1970**]; [**Gaito1980**]), it remains routine in survey-based social science. In political science and sociology alike, researchers frequently correlate 4–, 5–, or 7–point items as if they were interval-scaled, usually without examining the implications of the marginals. Examples include correlations among attitude items in large-scale survey batteries ([**AlwinKrosnick1991**]), ideology and issue positions on mixed-resolution scales ([**BrewerGross2005**]; [**KinderKam2009**]), and summary indices that are subsequently modeled using correlation-based techniques. Reviews document that such practices are pervasive in applied survey analysis ([**KreuterValliant2007**]). Methodological discussions sometimes defend pragmatic use of parametric tools with ordinal data ([**Jacoby1999**]), but even optimistic views do not address the *feasible range* of r imposed by fixed marginals—an omission with direct consequences for substantive interpretation.

Alternatives to Pearson's r—notably Spearman's ρ, Kendall's τ, and the polychoric correlation—offer different trade-offs ([**Spearman1904**]; [**Kendall1938**]; [**Olsson1979**]). Rank-based measures ignore the marginals; polychorics assume latent bivariate normality. In contrast, we pursue bounds on r that *explicitly* depend on the observed marginals, drawing on rearrangement inequalities and the Fréchet–Hoeffding framework ([**Frechet1951**]; [**Hoeffding1940**]; [**Rueschendorf1981**]). This perspective clarifies when the attainable range of r is narrow, when symmetry fails ($r_{\min} \neq -r_{\max}$), and when the lower bound can be strictly positive. It also explains why correlation-based models with ordinal indicators (EFA/CFA/SEM) can be distorted when Pearson correlations are used ([**FloraCurran2004**]; [**GreenEtAl1997**]). Our contribution situates empirical practice within these theoretical limits and provides tools to compute and visualize the bounds given fixed marginals.

This article proceeds as follows. We start by briefly elaborating the character of ordinal categorical variables (orcats), why they are generally not regarded as being metric, and why, in spite of that, they are nevertheless very frequently used in statistical procedures that assume metric level data. We then proceed in section 3 to show why correlations based on orcats tend to be constrained, which implies that their range is smaller than from -1 to +1, and why this is problematic in applied research. This raises the question how to determine the actual range of attainable values of correlations between orcats, which we address in the subsequent section. In section 5 we assess the magnitude of these constraining effects on two different sets of data. One is a typical survey dataset in which many of the variables are ordinal categorical, the orcatorcat 2019 British Election Study. The second consists of data derived from simulations covering all possible relevant conditions for pairs of ordered categorical variables. We use both sets of data to explore conditions that influence the magnitude of the constraining effect. Section Section **??** concludes by summarizing the findings, by elaborating the consequences for applied research, and by discussing directions for further research.

1.1 On ordinal-categorical variables (orcats), their level of measurement and their common use in quantitative analyses

Orcats are ordinal variables with a relatively small number of categories (or ranks) compared to the number of cases, thus resulting in partial orderings of cases. Although they can be found in all field of quantitative inquiry they are ubiquitous in data from standardised social science surveys. Likert items and rating scales [describe in fn Likert items and rating scales, add that there are yet other formats such as semantic-differential scales that also yield orcats] are among the most favoured ways to elicit responses in such surveys. These survey-question formats have most often between 3 and 11 ordinal categories. The numerical values assigned to the categories of orcats are most often consecutive integers, starting with either 0 or 1. [fn: explain that these values may be included in the formulation of a survey question, which is immaterial to our discussion here. The values assigned to the categories (and to the responses) reflect, in combination with the analysis procedure the analyst's interpretation of their meaning.] . Such an assignment of values is as good as, and indeed analytically equivalent to, any other set of values that reflects the analyst's understanding of the ordinal relations between the categories, i.e., any monotonic transformation. However, objections are often raised when orcats [fn: add references] are analysed with procedures requiring metric measurement. That requirement implies that the differences between values can validly be compared so that, e.g., the substantive difference between categories 1 and 2 is equal to that between categories 3 and 4, and half as large as the difference between categories 2 and 4, etc. Any implied interval-level interpretation of ordinal categorical variables requires therefore a compelling theoretical or empirical justification of the numerical values assigned to the categories rather than just the analyst's fiat. In principle this is quite possible, for example by calibrating values of categories against psychophysical measurements (cf., Lodge; Tillie; others), or against the results of psychometric/linguistic analysis of verbal response labels (cf Sönning 2024). However, such procedures are rarely integrated in applied survey research because of their high cost. In the absence of compelling justifications for values assigned to categories orcats can only be considered to provide ordinal information, not metric information. In spite of this we find that they are very commonly analysed with statistical tools that require metric measurement of variables such as product-moment correlations or procedures that build upon such correlations. We find this clearly exhibited in published articles in high-ranking academic journals [fn: explain how we did this] and in highly regarded textbooks where examples are given of the use of correlations between orcats as input for procedures such as PCA, FA or SEM [fn: examples: Kline; Byrne; xxx]. This practice is so established that it is occasionally rationalised by referring to orcats as being 'quasi-interval level' measures [fn: reference]. One can, however, also find more explicit justifications for this practice, which emphasise the robustness of correlation coefficients for violations of their measurement level assumptions. Labovitz (1967, 1970) in particular has compared correlations between orcats scored according to equal-distances with correlations when the variables were scored with a monotonic random procedure (which only assumes ordinality between categories). He reports that the differences are negligible, and suggests that even if the theoretical objection against unfounded interval level interpretations of orcats would be justified, that objection would nevertheless have such minor practical consequences as to make it irrelevant in most situations. Not surprisingly, Labovitz' work has in turn be criticised itself on various grounds, most particularly of (inadvertently) mainly covering situations where this robustness holds, and overlooking other situations (cf O'Brien 1979; xxx). At this place we will not focus on discussions about the equal-distance scoring practice. Our main reason for referring to them is to demonstrate the existence of lively debates in the applied literature about potential issues when using orcats in procedures requiring metric data. The existence of these debates makes it remarkable that another important issue has –to the best of our knowledge– remained mainly overlooked in the applied literature while it has been well-known in the theoretical statistical literature. This is the problem that correlations between orcats are generally 'constrained', which means that they do not range between -1 and +1, but between minima and maxima that have to be calculated separately for each pair of variables. The nature of this issue is the subject of the next section.

1.2 Bounds of correlations between ordinal categorical variables (orcats)

The values that product-moment correlations between ordinal-categorical variables (orcats) can attain are generally constrained by minimum and maximum bounds of attainable values that are not -1 and +1. [fn: The maximum attainable value is +1 only when the univariate distributions of the variables are identical. But that does not imply the minimum attainable value then to be -1, as that also requires further conditions that the univariate distributions ,must meet (see below, in the discussion of Table 2). The minimum attainable value of a correlation is -1 when the univariate distributions are each other's inverse, but again, that does not imply that the maximum is +1, as that requires additional conditions of the univariate distributions.]

Because the bounds of correlations between orcats are not -1 and +1 it is even more difficult than commonly thought in applied research to interpret the strength of correlations. The well-known and often referred to criteria formulated by Cohen (1988: 77-81) for correlations in the behavioural sciences [fn: Cohen suggested, with appropriate caution, the following guidelines: $0.10 < r < 0.29$ weak; $0.30 < r < 0.49$ moderate; $0.50 < r < 1.00$ strong. Hemphill (2003) suggests on the basis of a review of several meta-analyses in psychological research that the cut-off of 0.50 for considering a correlation to be 'strong' may be too high in many common instances.] are explicitly based on the notion that the minimum and maximum attainable values for r are -1 and +1. But that is generally not the case and, moreover, the actual upper- and lower-bounds of r are not the same across pairs of variables, but specific to each. This state of affairs thus generates two interconnected problems when correlating orcats. First, the absence of clarity of the lower- and upper-bounds of r makes qualitative interpretations of the strength of correlations elusive. Second, because the bounds of attainable values of r are specific for each pair of variables each correlation is expressed in terms of a different, but unknown reference interval. Such correlations are therefore not comparable. To better grasp these problems, it is necessary to determine the upper- and lower bounds of correlations, given the univariate distributions of the variables involved.

Statistical scholarship has since long recognized that, in contrast to what is ofen thought, the upper- and lower-bounds of correlations between orcats are generally not the reference interval [-1, +1]. [insert references, including Shi&Huang 1992; xxxx]. Determining the actual bounds given the marginal distributions of the two variables can be done by using Whitt's (1976) rearrangement theorem and Hardy–Littlewood–Pólya's (1952) rearrangement inequality. For obtaining the maximum attainable value of a correlation the values of X and Y have each to be sorted independently in either ascending or descending order and then paired (comonotonic arrangement). For the minimum attainable value of the correlation (given the marginals of the two variables) the anti-comonotonic arrangement is required (sorting one of the variables in ascending and the other in descending order). A formalized elaboration is presented in Appendix 1. Moreover, the algorithm for calculating rmin and rmax has been programmed in R and is [or: will be] freely available for researchers to assess the minimum and maximum attainable values of r for pairs of variables in their own data. Details about this routine are described in Appendix 2.

A series of illustrative cases is presented in Table 2 (as in Table 1 for n=12 and two orcats with 4 categories each). Although the distributions of the variables given in these examples is of no particular importance, and could have been replaced by innumerable other examples, they are nevertheless helpful to arrive at some general observations about rmin and rmax.:

- The values of rmin and rmax are not constant across all examples, and not even across correlations involving a specific variable (see examples (a) and (b) which involve the same X-variable). The variation in the values of the bounds of r is across pairs of variables. In other words, the bounds of r are specific for each pair.
- rmin max can be -1, and rmax can be 1, but neither of these values implies the other (as seen in examples (b) and (c). The combination of rmin=-1 and rmax=1 is possible, but dependent on specific characteristics of the two marginal distributions involved (see examples (d) and (f). Both variables must not only have identical distributions, but these must also be symmetrical.

- rmin and rmax are not necessarily symmetric around 0. Such symmetry may exist, but requires specific characteristics of the pair of marginal distributions, namely that both are symmetric (see examples (e), (g) and (h).

- The extent to which the reference interval [-1, 1] is covered by the interval [rmin, rmax] varies across pairs of variables.
- The values of rmin and rmax depend not only on the marginal distributions of the two variables, but also on the values assigned to their categories. Unless specified differently, we follow in this article the common practice of assigning consecutive integer values to the ordered categories. But any monotonic transformation of values is equally adequate to represent the ordinal character of variables. Examples (i) and (j) involve the same pair of variables, but in (i) their categories have been assigned values 1. 2, 3 and 4, while in (j) the assigned values were 1, 2.25, 2.75 and 4. The consequences of this change are visible in the values of rmin and rmax. This observation implies that value assignment to categories of orcats (and the justification thereof) remains an issue of importance when using Pearson correlations on orcats. It is a different problem, however, than that of constrainment of correlations on which we focus here.

Consequences of constrained correlations in applied research

- Descriptive interpretation of correlations (Cohen's criteria) is undermined
- It is also undermined by every correlation being expressed on its own unique scale/unique calibration
- Need a nice example of 2 correlations with $r(a,b) \leq r(b,c)$ and where $r(a,b)$max is $< r(b.c)$: is it smaller because the linear relationship is weaker, or because he constrainment is stronger?

1.3 Background and Definitions

- Ordinal data and assumptions of interval-scale correlation

- Existing alternatives: Spearman, polychoric, polyserial; their strengths and limitations

- Coupling and rearrangement in discrete settings

2 Theoretical Framework

The problem of extremal correlations under fixed marginals connects directly to classical results in probability inequalities. Fréchet's work on admissible joint distributions ([**frech51-sur**]) and Hoeffding's development of covariance bounds ([**hoef40-massta**]) laid the groundwork. Later refinements, such as Rüschendorf's demonstration of sharpness of Fréchet bounds ([**rues81-sharpn**]), clarified when comonotonic and countermonotonic couplings achieve extremal values. The rearrangement inequality, and its discrete extensions, ensure that matching high values of X with high values of Y maximizes $\mathbb{E}[XY]$, while opposite pairings minimize it. This perspective is elaborated in comparison and coupling frameworks ([**muel02-compa**]; [**thor00-coupli**]). These results provide the theoretical backbone for our derivations of $r_{\min}$ and $r_{\max}$.

2.1 Optimal Coupling and Correlation Bounds

We formalize the problem of identifying the joint distribution (coupling) of two ordinal variables with fixed marginals that maximizes or minimizes the Pearson correlation.

Proposition 2.1. *Let X and Y be discrete ordinal random variables, each with finite ordered support:*

$$X : x_1 < x_2 < \cdots < x_{K_X}, \quad Y : y_1 < y_2 < \cdots < y_{K_Y}$$

with marginal probability distributions $p_X(x_i) = P(X = x_i)$, $p_Y(y_j) = P(Y = y_j)$, respectively.

Then, the maximum and minimum values of the Pearson correlation coefficient r_{XY}, consistent with the fixed marginals, are obtained as follows:

- Maximum r_{XY}: achieved by pairing largest X-values with largest Y-values (comonotonic arrangement).

- Minimum r_{XY}: achieved by pairing largest X-values with smallest Y-values (anti-comonotonic arrangement).

Proof. The Pearson correlation coefficient is given by:

$$r_{XY} = \frac{E[XY] - E[X]E[Y]}{\sigma_X \sigma_Y}.$$

Given fixed marginals $p_X(x_i)$, $p_Y(y_j)$, the expectations $E[X], E[Y]$ and standard deviations σ_X, σ_Y are constant. Thus, to maximize or minimize r_{XY}, we only need to maximize or minimize the expectation $E[XY]$:

$$E[XY] = \sum_{i=1}^{K_X} \sum_{j=1}^{K_Y} x_i y_j P(X = x_i, Y = y_j).$$

This proof relies on a rearrangement inequality of [**hard52-inequalities**] (Theorem 368):

For real sequences $a_1 \leq a_2 \leq \cdots \leq a_n$ and $b_1 \leq b_2 \leq \cdots \leq b_n$, and for any permutation π of indices $\{1, \ldots, n\}$ we have the following:

$$a_1 b_1 + a_2 b_2 + \cdots + a_n b_n \geq a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \cdots + a_n b_{\pi(n)}$$

and

$$a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1 \leq a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \dots + a_n b_{\pi(n)}$$

Equality occurs only if the permutation π results in a monotonically increasing sequence (for maximum) or monotonically decreasing sequence (for minimum).

To explicitly connect this to our problem, enumerate observations explicitly from sorted marginals:

$$X_{\downarrow} : x_{[K_X]} \geq x_{[K_X-1]} \geq \dots \geq x_{[1]}, \quad Y_{\downarrow} : y_{[K_Y]} \geq y_{[K_Y-1]} \geq \dots \geq y_{[1]}.$$

(Sorted in descending order for maximization). Next expand discrete probability distributions into explicit observation-level sequences. If $K_X \neq K_Y$, extend the shorter sequence with zero-probability categories so both sequences have equal length, which does not affect marginal probabilities or correlation.

Hence, to maximize $E[XY]$: pair the largest ranks of X with the largest ranks of Y. Explicitly, start from the top (highest values) and move downward:

- Pair as many observations of the largest X-category as possible with the largest Y-category, then next-largest, etc., until all observations are paired.
- By the Hardy–Littlewood–Pólya Rearrangement Inequality [**hard52-inequalities**], this explicitly constructed pairing yields the maximum possible sum of products, hence maximizing $E[XY]$.

Likewise, to minimize $E[XY]$ pair the largest X-categories with the smallest Y-categories, then second-largest X-categories with second-smallest Y-categories, and so forth (anti-comonotonic ordering). By the rearrangement inequality, this arrangement explicitly yields the minimal sum of products, hence minimizing $E[XY]$. ■

In words, given fixed marginals, we have shown:

- **Maximum correlation** $r_{\max}$ is achieved by comonotonic (e.g. sorting values of X and Y from largest to smallest independently and pairing) arrangement:

$$P_{\max}(X = x_{[i]}, Y = y_{[i]}) \quad \text{in either descending or ascending order.}$$

- **Minimum correlation** $r_{\min}$ is achieved by anti-comonotonic arrangement:

$$P_{\min}(X = x_{[i]}, Y = y_{[n+1-i]}) \quad \text{pairing descending X with ascending Y, or viceversa.}$$

With this explicit construction we can simply calculate r_{max} and r_{min} when the values of X and Y are ranks.

Corollary 1. (Analytic Forms for $r_{\min}$ and $r_{\max}$)

Let two ordinal variables X and Y have values as follows:

- X with K_X ordinal levels: $1, 2, \dots, K_X$

- Y with K_Y ordinal levels: $1, 2, \dots, K_Y$

with absolute frequencies:

- $n_X(i)$, number of observations at level i of X.

- $n_Y(j)$, number of observations at level j of Y.

Then the maximum Pearson correlation is:

$$r_{\max} = \frac{E[XY]_{\max} - \bar{X}\bar{Y}}{\sigma_X \sigma_Y}.$$

Where

$$E[XY]_{\max} = \frac{1}{N} \sum_{i=1}^{K_X} \sum_{j=1}^{K_Y} x_i y_j M_{i,j}$$

where explicitly defined frequencies $M_{i,j}$ pair observations in a comonotonic manner, calculated explicitly via the greedy iterative algorithm in as Algorithm **??** in the proof.

Then explicitly the minimal expectation is:

$$E[XY]_{\min} = \frac{1}{N} \sum_{i=1}^{K_X} \sum_{j=1}^{K_Y} x_i y_j R_{i,j}$$

Where R_{ij} is defined in the proof.

Proof. Algorithm for explicit calculation of $M_{i,j}$ (greedy method):

Initialize available frequencies explicitly: $n'_Y(j) = n_Y(j) \quad \text{for each } j$

- For each level x_i, from smallest $i = 1$ to largest:
  - For each level y_j, from smallest $j = 1$ to largest:
    - Pair as many observations as possible:

$$M_{i,j} = \min\left(n_X(i) - \sum_{s=1}^{j-1} M_{i,s},\, n'_Y(j)\right)$$

Next, update explicitly:

$$n'_Y(j) \leftarrow n'_Y(j) - M_{i,j}$$

Define the matrix $M_{i,j}$, the frequency with which the level x_i pairs with the level y_j. Clearly, we must have:

$$\sum_j M_{i,j} = n_X(i), \quad \sum_i M_{i,j} = n_Y(j)$$

Similarly, the minimum Pearson correlation (r_{min}) achieved by anti-comonotonic alignment (pairing largest ranks of X with smallest ranks of Y) is explicitly:

$$r_{\text{min}} = \frac{E[XY]_{\text{min}} - \bar{X}\bar{Y}}{\sigma_X \sigma_Y}.$$

To achieve the maximum expected product $E[XY]_{\text{max}}$, do the following: First sort ranks from largest to smallest independently for both X and Y; then pair ranks directly from highest to lowest available observations.

Minimum Expectation

The formula for $E[XY]_{\text{min}}$ is derived in a similar manner. Define explicitly a matrix $R_{i,j}$ giving frequencies of pairings $X = x_i$ with $Y = y_j$. For the minimal expectation, pair largest available X-values explicitly with smallest available Y-values first (greedy algorithm):

Initialize explicitly available frequencies:

$$n'_Y(j) = n_Y(j) \quad \text{for each } j$$

- For i from largest to smallest ($i = K_X$ down to 1):
  - For j from smallest to largest ($j = 1$ up to K_Y):
    * Pair explicitly as many observations as possible:

$$R_{i,j} = \min\left(n_X(i) - \sum_{s=1}^{j-1} R_{i,s},\, n'_Y(j)\right)$$

- Update explicitly available frequencies:

$$n'_Y(j) \leftarrow n'_Y(j) - R_{i,j}$$

Then explicitly the minimal expectation is:

$$E[XY]_{\text{min}} = \frac{1}{N} \sum_{i=1}^{K_X} \sum_{j=1}^{K_Y} x_i y_j R_{i,j}$$

This explicit equation applies Whitt's rearrangement theorem , **whit76-bivariate** , ensuring proper maximal pairing of ranks. (**whit76-bivariate**) ■

Discussion

We want to construct a joint distribution table M such that:

- The **row sums** match the marginal counts for X: $\sum_j M_{i,j} = n_X(i)$
- The **column sums** match the marginal counts for Y: $\sum_i M_{i,j} = n_Y(j)$

To do this, we go **greedily**, pairing the highest available X values with the **smallest available** Y values, one cell at a time.

To avoid **exceeding** the available marginal totals in Y, we must track how much of $n_Y(j)$ is **still unassigned** at each step — and that's what $n'_Y(j)$ represents.
After assigning $M_{i,j}$ units to the cell (i,j), we subtract that amount from the remaining pool of level y_j values, like this: $n'_Y(j) \leftarrow n'_Y(j) - M_{i,j}$This ensures we don't **overfill** any column of the matrix.

To maximize the expectation $E[XY]$, the rearrangement inequality states we must pair values of X and Y from smallest-to-smallest, second-smallest-to-second-smallest, and so forth (comonotonic alignment).

Thus, explicitly:

- Sort X: lowest-to-highest:

$$X_{\text{sorted}} = (\underbrace{x_1, \dots, x_1}_{n_X(1)}, \underbrace{x_2, \dots, x_2}_{n_X(2)}, \dots, \underbrace{x_{K_X}, \dots, x_{K_X}}_{n_X(K_X)})$$

Sorted Y-values (ascending order for max, descending for min):

$$Y_{\text{sorted(max)}} = (\underbrace{y_1, \dots, y_1}_{n_Y(1)}, \underbrace{y_2, \dots, y_2}_{n_Y(2)}, \dots, \underbrace{y_{K_Y}, \dots, y_{K_Y}}_{n_Y(K_Y)})$$

$$Y_{\text{sorted(min)}} = (\underbrace{y_{K_Y}, \dots, y_{K_Y}}_{n_Y(K_Y)}, \underbrace{y_{K_Y-1}, \dots, y_{K_Y-1}}_{n_Y(K_Y-1)}, \dots, \underbrace{y_1, \dots, y_1}_{n_Y(1)})$$

The maximum and minimum expectations are simply:

Maximum expectation (pair element-by-element):

$$E[XY]_{\max} = \frac{1}{N} \sum_{k=1}^{N} X_{\text{sorted}}(k) \cdot Y_{\text{sorted(max)}}(k)$$

- **Minimum expectation**:

$$E[XY]_{\min} = \frac{1}{N} \sum_{k=1}^{N} X_{\text{sorted}}(k) \cdot Y_{\text{sorted(min)}}(k)$$

These classical bounds define feasible joint distributions given marginals and constrain the attainable values of r. There is an interesting connection to Fréchet–Hoeffding and Boole–Fréchet Inequalities, which we note but leave for future research.[1]

2.1.1 Illustration

To illustrate these idea concretely, Table 1 presents an illustrative example of comonotonic and anti-comonotonic arrangements and of the resulting lower- and upper-bounds of r, for two 4-category variables and 12 cases. One can easily expand this example to pairs of variables with different numbers of categories and different numbers of cases.

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[1]Fréchet–Hoeffding and Boole–Fréchet Inequalities: The upper bound is always attained by r_{max}. However, the lower bound is *not* guaerenteed to attain. The fundamental asymmetry arises because discrete ordinal distributions cannot smoothly bridge these gaps. Thus, the upper bound alignment is always feasible, while the lower bound alignment frequently encounters structural obstacles.

In the discrete ordinal setting, $r_{\max}$ is **always** attained by the FH upper (comonotonic) construction. *(BFbounds.md; Proofs-OrdinalVariables.md; Ordinal Variables and Correlation.md)*

The FH lower (countermonotonic) construction may be **unattainable** in discrete settings; $r_{\min}$ is then the best feasible opposite-order arrangement.
(BFbounds.md; Proofs-OrdinalVariables.md; Ordinal Variables and Correlation.md)

Upper bound (positive alignment):

- Sorting X and Y separately in descending order, then pairing largest-to-largest, smallest-to-smallest, **always works**.
- You never run out of "matching" observations because the descending order for both variables naturally aligns category sizes. Even if one variable has large gaps, you can still sequentially pair without difficulty.

Lower bound (negative alignment):

- You sort X descending but Y ascending. You're pairing extremes together: highest X with lowest Y, lowest X with highest Y.
- Now you face potential "imbalance":
  - If one category is large for X but the corresponding opposite-ranked category for Y is small, you quickly exhaust available "opposite-end" observations.
  - After exhausting the best opposites, you are forced to pair less "extreme" pairs, creating a suboptimal alignment.
- Hence, perfect negative alignment (lower bound) becomes impossible to achieve fully, leaving you short of the theoretical Fréchet–Hoeffding lower bound.

Table 1: Determining minimum and maximum attainable values of correlation between two ordered categorical variables X and Y given their marginal distributions.

(a) Observed frequency distributions

	Category 1	**Category 2**	**Category 3**	**Category 4**	**Total**
X	6	3	2	1	12
Y	3	5	2	2	12

(b) Comonotonic arrangement and pairing:

X: 1 1 1 1 1 1 2 2 2 3 3 4

Y: 1 1 1 2 2 2 2 2 3 3 4 4 $r_{\text{max}} = 0.87831$

(c) Anti-comonotonic arrangement and pairing:

X: 1 1 1 1 1 1 2 2 2 3 3 4

Y: 4 4 3 3 2 2 2 2 2 1 1 1 $r_{\text{min}} = -0.79466$

Table 2: Illustrative pairs of marginal distributions and their associated minimum and maximum attainable values of r ($r_{\min}$, $r_{\max}$) given these distributions. The distributions are given in counts (n=12). The categories are assigned consecutive integer values starting with 1. C1 and C2 are measures of constraint, as discussed in the text.

	Category				**Total**	$r_{\min}$	$r_{\max}$
	1	**2**	**3**	**4**			
(a) Two arbitrarily distributed ordinal categorical variables (orcats)							
X	6	3	2	1	12	-0.79466	0.87831
Y	3	5	2	2	12		
(b) One arbitrarily distributed variable, and one with same distribution inversed							
X	6	3	2	1	12	-1	0.71429
Y	1	2	3	6	12		
(c) Two variables with arbitrary identical distributions							
X	6	3	2	1	12	-0.71429	1
Y	6	3	2	1	12		
(d) Two variables with identical symmetric and unimodal distributions							
X	2	4	4	2	12	-1	1
Y	2	4	4	2	12		
(e) Two variables with different distributions, both symmetric and unimodal							
X	2	4	4	2	12	-0.91169	0.91169
Y	1	5	5	1	12		
(f) Two variables with identical symmetric bimodal distributions							
X	4	2	2	4	12	-1	1
Y	4	2	2	4	12		
(g) Two variables with different distributions, both symmetric and bimodal							
X	4	2	2	4	12	-0.95673	0.95673
Y	5	1	1	5	12		
(h) One symmetric unimodal variable, the other symmetric bimodal							
X	2	4	4	2	12	-0.89923	0.89923
Y	4	2	2	4	12		
(i) Two variables, one very left skewed, the other very right skewed							
X	1	1	1	9	12	-0.95818	0.31939
Y	8	2	1	1	12		
(j) Same as situation (i), above, but categories assigned values 1; 2.25; 2.75; 4							
X	1	1	1	9	12	-0.93321	0.33829
Y	8	2	1	1	12		

2.2 Examples

Below, "scores 0:$(K-1)$" means category values $[0, 1, \ldots, K-1]$. "Affine recode" means $a + b \times$ (old score) with $b > 0$. "Non-affine monotone" means strictly increasing but not affine (e.g., convex spacing).

2.2.1 Notation (permutation extremes, "Lane A")

- Build expanded vectors x and y from **counts** and **scores**.
- $r_{\max} = \mathrm{cor}(\mathrm{sort}(x, \downarrow), \mathrm{sort}(y, \downarrow))$.
- $r_{\min} = \mathrm{cor}(\mathrm{sort}(x, \downarrow), \mathrm{sort}(y, \uparrow))$.

2.2.2 A) Same number of categories, but $|r_{\min}| \neq r_{\max}$ (concrete numbers)

Counts (both

$$K = 5$$

):

- $$X: \ [100,\ 200,\ 300,\ 200,\ 200]$$
- $Y: \ [100,\ 100,\ 100,\ 300,\ 400]$ (skewed; not palindromic)

Scores: 0:4 for both.

- $$r_{\max} \approx 0.929398758$$
- $$r_{\min} \approx -0.881118303$$
- $$|r_{\min}| \approx 0.8811 \neq r_{\max}$$

Affine recode (10,20,30,40,50) for both:

- $r_{\max}, r_{\min}$ **unchanged** (up to rounding).

Non-affine monotone recode for Y: $[0, 1, 1.5, 3, 10]$.

- $$r_{\max} \approx 0.930261031$$
- $$r_{\min} \approx -0.863250703$$
- Values change (Pearson is **not** invariant to non-affine spacing); magnitudes still unequal.

Why unequal magnitudes? No palindromic symmetry in Y the "palindrome-sum" pattern in y'−space isn't zero, so $S_{\max} + S_{\min} \neq 0$.

2.2.3 B) Different numbers of categories, yet $|r_{\min}| = r_{\max}$ (concrete numbers)

Counts:

- X (4 categories): [100, 200, 300, 400]
- Y (5 categories): [100, 200, 400, 200, 100] (**palindromic**)

Scores: $X:$ 0:3, $Y:$ 0:4.

- $$r_{\max} \approx 0.912870929$$
- $$r_{\min} \approx -0.912870929$$
- $|r_{\min}| = r_{\max}$ (to machine precision)

Affine recode (e.g., 10,20,...):

- Magnitudes remain **identical**; affine transforms don't change Pearson correlation.

Non-affine monotone recode for Y: $[0, 1, 1.5, 3, 10]$.

- $$r_{\max} \approx 0.580072921$$
- $$r_{\min} \approx -0.842041337$$
- **Magnitudes no longer match** because antisymmetry of the **scores** about the center is lost, even though the **counts** are palindromic.

Why equal magnitudes in the affine case? Palindromic **mass** symmetry in Y + **antisymmetric scoring** (any affine map of 0:$(K_Y - 1)$) the palindrome-sum vector $y'_i + y'_{n+1-i} \equiv 0$. Hence $S_{\max} + S_{\min} = 0$ $|r_{\min}| = r_{\max}$ for **any** X, irrespective of $K_X \neq K_Y$.

2.2.4 C) Spearman invariance (same data as in B)

Compute ranks (with average ties) **once** for x and y, then form extremes by sorting those rank vectors.

- With Y scored 0:4:
  $r^{\text{Spearman}}_{\max} \approx 0.920837215, \quad r^{\text{Spearman}}_{\min} \approx -0.920837215.$
- With Y scored non-affinely $[0, 1, 1.5, 3, 10]$:
  identical Spearman extremes: 0.920837215 and -0.920837215.

Spearman is invariant to any strictly monotone recode because ranks depend only on order, not spacing.